

No-Logging as Code (the privacy promise made mechanical)

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Master technical specification — Volume 5 (Security & Privacy), nano-detail spec. Deepens invariant **S6** (no durable connection/traffic log — by construction) and **S7** (control actions audited; traffic never) of [04-sec] (04-security-privacy-pki.md §6–§7). This document owns the *mechanical contract* that turns "we don't log" from an operator's good intentions into a **build property**: the data-minimization model (what is stored vs never-stored), the CI/pre-build **schema-lint** that fails the build if any log/event schema acquires a traffic/PII shape, the ephemeral-only presence model (Redis TTL, no durable connection records), the control-action-only audit model, the §11.4.10 credentials discipline, and the paired §1.1 mutation that proves the lint is not a tautology. This is a **SPEC** — it describes the implementation; it does not build the product.

Source evidence cited inline by id: [04-sec §6/§7] (04-security-privacy-pki.md no-logging-as-code + audit), [svc-tel] (v03-control-plane/svc-telemetry.md — the telemetry service that *governs* the schema-lint + owns presence + audit-sink), [svc-evt] (v03-control-plane/svc-events.md — the event bus C3 payload constraint + payload-lint), [04_P1 §11.4] (HelixVPN-Phase1-MVP no-logging CI lint), [04_ARCH §7/§4.5] (no-logging-as-code

- ephemeral Redis presence). Access date for external facts: **2026-06-25**. Unproven claims are marked UNVERIFIED: per §11.4.6 — never fabricated.

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0. Position, ownership, and the one-sentence contract

One-sentence contract. HelixVPN's no-logging guarantee is enforced by *absence* and by a CI/pre-build **schema-lint** that fails the build if any durable table, any event payload, or any audit blob acquires a connection/traffic/PII shape — so the privacy promise is a mechanical property of the build, not an operator's good intentions [04-sec §6, 04_P1 §11.4, svc-tel §7].

What this document owns:

- the data-minimization model (the closed stored-vs-never-stored partition);
- the forbidden-field denylist + the topology/identity allow-list;
- the schema-lint gate (static migrations + live `information_schema`) + its verdict logic;
- the event-payload lint (C3 on the bus) and the audit meta-shape guard;
- the paired §1.1 mutation that proves the lint catches a planted flow table;
- the runtime-signature requirement (green against the *deployed* DB, §11.4.108);
- the §11.4.10 credentials intersection and the GDPR posture.

What this document does NOT own (referenced, not redefined):

- the canonical DDL for `audit_events`, `devices`, `advertised_prefixes` — that is the data model (doc 02 / svc-tel); this doc quotes the relevant slices and the lint over them;
- the telemetry service's presence/audit/metric *implementation* — that is [svc-tel]; this doc owns the *invariant* (S6/S7) the telemetry service mechanizes;
- the event bus's transport/consumer machinery — that is [svc-evt]; this doc owns the *C3 payload-shape constraint* on the bus, not the bus itself.

Governance note (§11.4.35). The schema-lint binary lives at `tools/schemalint` but is **governed by the telemetry service** because "the privacy guarantee is telemetry's charter" [svc-tel §0]. This document is the cross-volume security spec for the *invariant*; [svc-tel §7] is the service-level nano-detail of the lint's parser. They agree by construction.

1. The privacy promise as a build property, not a config flag

A privacy VPN that *could* log but promises not to is one config edit, one rogue operator, or one well-meaning "let's add a debug table" away from betraying every user. HelixVPN closes this class **by construction**: the data simply does not exist, and a build gate refuses any change that would make it exist [04-sec §6, 04_ARCH §7].

Three cooperating mechanisms make the promise mechanical:

1. **Absence by design (S6)**. There is no connections, flows, traffic, packets, or durable sessions table anywhere in the schema. Per-flow / per-packet / per-destination data is *never written* — not "written then redacted," **never written** [04-sec §0.1 S6, §6.1].
2. **The schema-lint gate (§6)**. A pre-build + post-deploy lint introspects *both* the migration SQL *and* the live `information_schema` and **fails the build** if any durable table is connection-log-shaped (forbidden name, or a src/dst column + a bytes/timestamp column outside `audit_events`) [04-sec §6.2, svc-tel §7]. The privacy promise is enforced by tooling, not trust.
3. **Ephemeral-only live state (§3)**. "Who is online right now" lives only in Redis with a TTL, so the durable store never accumulates a connection log [04-sec §6.1, 04_ARCH §4.5, svc-tel §5]. Online/offline is operationalized *without* a session table.

The bar (per the §11.4 covenant): a green build is the *guarantee* the data cannot exist, backed by a paired §1.1 mutation (§10) proving the lint is not a tautology that passes everything.

2. The data-minimization model — stored vs never-stored

The single most important table in this document: the **closed partition** of what HelixVPN's durable store holds vs what it can never hold [04-sec §6.1, svc-tel §5].

Stored (durable, Postgres)	NEVER stored (no table, no column, ever)
Identity : tenants, users (<code>oidc_sub</code> /email only for managed tenants), devices (<code>wg_pubkey</code> , <code>overlay_ip</code> , coarse <code>last_seen_at</code>)	Per-connection records (one row per tunnel session)
Topology : connectors, advertised_prefixes (CIDRs), groups, sites	Per-packet / per-flow records (src IP, dst IP, ports-in-use, byte counts, timestamps-of-use)
Policy : policy spec + compiled rules + version history	DNS query logs (which domains a user resolved)

Stored (durable, Postgres)	NEVER stored (no table, no column, ever)
PKI: device_certs, enroll_tokens (Argon2id-hashed, never plaintext)	Destination IPs / domains visited by any user
Audit: audit_events (control actions only, §4)	Traffic content or metadata of any kind
Aggregate counters: handshakes, bytes-total, errors — Prometheus series, <i>not</i> per-user	Anything correlating a user to a destination or a timestamp-of-use

2.1 The two columns that look like traffic but are not (the allow-list)

Three identity/topology columns *do* carry IP/CIDR shape and are explicitly **allow-listed** — they are the device's *own* stable address or a connector's *advertised* network, not traffic [svc-tel §7.1, svc-evt §2.2]:

Column	Type	Why it is identity/topology, not traffic
devices.overlay_ip	inet	the device's own stable overlay address (its identity on the mesh), assigned at enroll — never a destination it talked to
devices.wg_pubkey	bytea	32-byte Curve25519 public key = the device identity
advertised_prefixes.cidr	cidr	a connector's <i>advertised</i> network (topology) — what the connector <i>offers</i> , not what anyone <i>visited</i>
overlay_pools.ula_prefix	cidr	IPAM — the tenant's ULA /48 allocation pool

The allow-list is asserted **exhaustive** (§6): any inet/cidr/*_ip/*_port-shaped column *not* in this list FAILs the lint. This is the discriminator that lets the lint pass legitimate identity/topology while failing any traffic-shaped addition.

2.2 The coarsening rule (last_seen_at cannot reconstruct a timeline)

devices.last_seen_at is the one durable timestamp derived from presence — and it is **deliberately coarse**. It is refreshed *at most* every LAST_SEEN_COARSEN (default **5 min**), **not** per-heartbeat, and carries **no destination** [04-sec §6.1, svc-tel §5.1]. A coarse 5-minute "last seen" cannot reconstruct a session timeline; it is enough for "is this device stale?" and nothing more.

3. Ephemeral-only presence (Redis TTL, no durable connection records)

The mechanism that lets HelixVPN know *which devices are online right now* **without a durable session/connection table** — the literal operationalization of the no-logging promise [04-sec §6.1, 04_ARCH §4.5/§4.6, svc-tel §5].

3.1 Presence is Redis-only, TTL'd, never copied to Postgres

Key	Type	Value	TTL	Set/refreshed by
presence: {tenant}: {device}	string	"1" or compact health JSON { "t": "masque- h3", "rtt": 21 }	PRESENCE_TTL (default 45 s)	MarkOnline/ RefreshPresence on each heartbeat
presence: {tenant}:set	set	device-ids currently online	members self- expire via the keyspace- notification reaper	reaper + Mark*

Three invariants make this the *operationalization* of no-logging [svc-tel §5.1]:

1. **Ephemeral by construction.** Presence lives only in Redis with a TTL = **3× the agent heartbeat interval** (PRESENCE_TTL = 3 * HEARTBEAT_INTERVAL, default 45 s = 3 × 15 s). The TTL is *derived* from the heartbeat cadence, never invented (§11.4.6). When a device stops beating, the key *expires* — there is no record it was ever there.
2. **Health-only value.** The value carries the *current* transport + rtt — **no destination, no bytes, no flow** (C3). Same restriction as the StatusReport proto message [svc-tel §5.1].
3. **Never copied to Postgres.** Presence is *never* persisted to the durable store. The only durable derivative is the coarse last_seen_at (§2.2). Losing Redis loses **no durable state** (C2) — and crucially, loses no privacy-sensitive data because none was durable.

3.2 Expiry → offline event (no silent corpus hole, §11.4.144 spirit)

```
stateDiagram-v2
    [*] --> Offline
    Offline --> Online: MarkOnline (WatchNetworkMap opened / first heartbeat)
    Online --> Online: RefreshPresence (heartbeat within TTL)
    Online --> Offline: MarkOffline (stream Close / explicit logout)
    Online --> Expiring: TTL elapsed, key evicted by Redis
    Expiring --> Offline: keyspace-notification reaper emits device.offline + SREM + gauge.Dec
    note right of Expiring
        A silently-expired presence key MUST still produce a
```

device.offline event
 (svc-tel §5.3) — the corpus-hole / fabricated-continuity
failure class (§11.4.144).
 But NOTHING about the session is written durably: offline
is an EVENT, not a row.
 end note

The expiry → offline event is mandatory so the coordinator flips relay availability — **but it is an ephemeral event on events: presence, not a durable record** [svc-tel §5.3]. A belt-and- suspenders sweeper (PRESENCE_SWEEP, default 30 s) reaps memberships whose key is gone, so even a missed keyspace notification never leaves a stuck "online" — and never writes a session row.

UNVERIFIED: Redis keyspace-expiry notifications are best-effort and can be delayed under load (Redis docs: expired events fire when the key is *accessed* or actively expired) [svc-tel §5.3]. Mitigation is the 30 s sweeper (the authoritative offline detector); notifications are the low-latency fast path. Neither path writes durable session data.

4. Audit — control actions only, traffic never (S7)

Every state-changing **control** action is audited; **traffic is never audited** because the traffic data does not exist (S6) — so it *cannot* be audited even by mistake [04-sec §7, svc-tel §4].

4.1 What is audited (the closed action vocabulary)

Audit covers **control actions only**, drawn from a closed Go enum so a typo cannot mint a new high-cardinality value and the action Prometheus label stays bounded [svc-tel §4.2, 04-sec §7]:

```
device.enrolled · device.revoke · device.cert.issued ·  
device.cert.rotated ·  
policy.create · policy.activate · policy.rollback ·  
connector.attached ·  
connector.prefixes.changed · enroll_token.mint · enroll_token.used  
·  
tenant.create · user.role.change
```

Audit() **rejects** any action not in this closed vocabulary with ErrUnknownAuditAction [svc-tel §4.2] — the mechanical guarantee that audit never silently grows toward traffic logging.

4.2 What is NEVER audited

- any packet, flow, destination, DNS query, or byte-count attributable to a user — **those don't exist (S6)**, so they cannot be audited [04-sec §7];
- the meta jsonb blob is shape-checked (§9) so traffic cannot be smuggled *inside* an audit row.

4.3 Audit is append-only by grant, control-only by vocabulary

The `audit_events` table is append-only by **grant**, not by trigger — the simplest mechanical guarantee [svc-tel §4.1]:

```
-- audit_events (canonical DDL in doc 02 / svc-tel §4.1; the no-
    logging-relevant constraints):
CREATE TABLE audit_events (
  id          bigint GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  tenant_id   uuid NOT NULL REFERENCES tenants(id) ON DELETE
              CASCADE,
  actor       text NOT NULL,                -- user id / "system"
              (never a destination)
  action      text NOT NULL,                -- closed vocabulary
              (§4.1)
  target      text,                         -- opaque id (device/
              policy/connector) – NOT a destination
  ts          timestampz NOT NULL DEFAULT now(), -- when the
              ADMIN acted, not a timestamp-of-use
  meta        jsonb                         -- small; shape-checked
              (§9) to carry no traffic shape
);
-- Append-only by grant: helix_app has INSERT + SELECT but NOT
  UPDATE/DELETE.
-- REVOKE UPDATE, DELETE ON audit_events FROM helix_app; -- a row
  cannot be mutated/erased by request-path.
```

`audit_events` is the **single allowed exception** in the schema-lint (§6): it is the only table permitted a `ts` column alongside `actor/target`, *because* it audits control actions — and even it is meta-shape-guarded (§9). Note the distinction the lint relies on: `audit_events.ts` is *when an admin acted*, never a *timestamp-of-use of traffic*; `target` is an opaque control-object id, never a destination.

5. The forbidden-field list (the closed denylist)

The schema-lint matches two regexes — a forbidden **table-name** set and a forbidden **column** set — against parsed SQL [svc-tel §7.1, 04-sec §6.2]. These are the concrete denylists.

5.1 Forbidden durable table names (connection-log shapes)

connections	connection	sessions	session	flows
flow	traffic			
packets	packet	netflow	bandwidth_samples	
dns_queries				
conn_log	access_log	session_log		

A durable table whose name matches `(?i)\b(connections?|sessions?|flows?|traffic|packets?|netflow| bandwidth_samples?|dns_queries?|conn_log|access_log|session_log)\b` **FAILs** the build [svc-tel §7.1]. The

name itself is a forbidden *signal* — privacy is about shape, and a `traffic_summary` view "over only counts" still FAILs (rename to `metrics_summary`; the lint defaults to FAIL = safe) [svc-tel §7.3].

5.2 Forbidden columns (traffic/PII shapes)

```
src_ip    dst_ip    dest_ip    src_port    dst_port
bytes_in  bytes_out                packet_count
payload   sni_host
flow_id   session_id  conn_id
```

A column matching `(?i)\b(src_ip|dst_ip|dest_ip|src_port|dst_port|bytes_(in|out)|packet_count|payload|sni_host|flow_id|session_id|conn_id)\b` **FAILs** [svc-tel §7.1]. Additionally, the *combination* rule from [04-sec §6.2]: any durable table (outside `audit_events`) that has **both** a src/dst-shaped column **AND** a bytes/ts-shaped column is a flow log by shape — FAIL, even if the individual column names dodge the explicit denylist.

5.3 The exhaustive allow-list (the ONLY permitted ip/port/cidr columns)

```
devices.overlay_ip          (inet)  – identity: the device's own
stable overlay addr (S6 not traffic)
devices.wg_pubkey           (bytea) – identity: 32B Curve25519
public key
advertised_prefixes.cidr    (cidr)  – topology: a connector's
advertised network
overlay_pools.ula_prefix    (cidr)  – IPAM: the tenant's ULA /48
pool
```

Any ip/port/cidr-shaped column **not** in this allow-list → FAIL [svc-tel §7.1]. The allow-list is asserted exhaustive: adding a new identity/topology column that carries IP shape requires an *explicit* allow-list amendment in the same change (a visible, reviewed decision per §11.4.6), not a silent pass.

6. The schema-lint gate (the mechanical enforcement)

The lint is the privacy guarantee turned into a build gate. It parses **both** the static migration SQL **and** the live `information_schema`, so a table that exists in the DB but not in a migration (drift) is also caught [04-sec §6.2, svc-tel §7].

6.1 What it parses and the verdict logic

```
// tools/schemalint/main.go (pre-build gate; constitution
                             §11.4.27 / §1.1 / §11.4.108 runtime sig)
// Parses BOTH migrations/*.sql (static) AND the live
   information_schema (dynamic) so DB drift
```



```
// (a table present in the DB but absent from migrations) is ALSO
// caught. SQL comments are
// stripped before matching (no false-fail on a prose mention of
// "bytes_out", svc-tel §7.3).
```

```
var forbiddenTables = regexp.MustCompile(
    `(?!i)\b(connections?|sessions?|flows?|traffic|packets?|
    netflow|bandwidth_samples?|` +
    `dns_queries?|conn_log|access_log|session_log)\b`)
var forbiddenCols = regexp.MustCompile(
    `(?!i)\b(src_ip|dst_ip|dest_ip|src_port|dst_port|bytes_(in|
    out)|packet_count|payload|` +
    `sni_host|flow_id|session_id|conn_id)\b`)
// allowList = { devices.overlay_ip, devices.wg_pubkey,
//               advertised_prefixes.cidr,
//               overlay_pools.ula_prefix } (asserted exhaustive,
//               §5.3)
```

```
flowchart TD
    START["pre-build gate runs schemalint"] --> PARSE["parse
migrations/ + live information_schema<br/>(strip SQL comments
first)"]
    PARSE --> T{"forbidden table-name match?"}
    T -- yes --> FAIL["exit 1: 'S6/C3 violation: durable traffic
table &name'"]
    T -- no --> C{"ip/port/cidr column outside the allow-list?"}
    C -- yes --> FAIL
    C -- no --> COMBO{"durable table (≠ audit_events) with src/dst
col AND bytes/ts col?"}
    COMBO -- yes --> FAIL
    COMBO -- no --> M{"audit_events.meta carries a forbidden-key
shape? (§9)"}
    M -- yes --> FAIL
    M -- no --> PASS["exit 0: no-logging invariant intact"]
    PASS --> BLOCK["build BLOCKED – release-blocker §11.4"]
```

6.2 The cross-check that closes the source-only-grep gap

A lint that only greps the migration *source text* is a §11.4.108 SOURCE-layer bluff: the deployed DB could have a drifted table no migration mentions. So the lint reads the **live information_schema** (the actual migrated schema, not the .sql text) [04-sec §6.2 "introspect the live migrated schema, not the .sql source"]. This makes the green verdict a **runtime signature** (§11), not a source-grep.

6.3 The four-test contract (§11.4.4(b) four-layer)

Layer	What	Where
pre-build gate	lint runs static + live parse; build BLOCKED on FAIL	tools/schemalint in the pre-build sweep
	lint runs against the <i>deployed</i> DB (§11)	

Layer	What	Where
post-deploy runtime signature		post-deploy assertion (§11.4.169 matrix)
paired §1.1 mutation	plant a flows(src,dst,bytes,ts) migration ⇒ lint MUST FAIL (§10)	meta-test
Challenge	end-to-end drive asserts schemalint green against the deployed DB	helix_qa/ challenges [svc-tel T-CHAL-1]

7. Schema examples — allowed vs rejected

Concrete cases the lint must classify correctly — **no false-pass, no false-fail** [svc-tel §7.3].

7.1 REJECTED (build BLOCKED)

```
-- REJECTED: forbidden table name + the canonical flow-log shape
               (this is the paired-mutation fixture, §10)
CREATE TABLE flows (
  src      inet,          -- forbidden: src-shaped
  dst      inet,          -- forbidden: dst-shaped
  bytes    bigint,        -- forbidden: bytes-shaped + (with src/
                        dst) the flow-log combination rule
  ts       timestampz     -- timestamp-of-use
);
-- → exit 1: "S6/C3 violation: durable traffic table flows"

-- REJECTED: dodges the table-name denylist but is a flow log by
               COMBINATION (src/dst + bytes/ts)
CREATE TABLE device_activity (
  device_id uuid,
  remote     inet,        -- dst-shaped (a destination the
                        device talked to)
  octets     bigint,      -- bytes-shaped
  seen_at    timestampz   -- ts-shaped
);
-- → exit 1: combination rule (src/dst col + bytes/ts col outside
               audit_events)

-- REJECTED: a forbidden column even on an otherwise-fine table
ALTER TABLE devices ADD COLUMN dst_ip inet; -- → exit 1:
               forbidden column dst_ip

-- REJECTED: a view whose NAME is a forbidden signal, even over
               only counts
CREATE VIEW traffic_summary AS SELECT count(*) FROM handshakes;
-- → exit 1: rename to metrics_summary
```

```

-- REJECTED: a connections-word column (aggregate must be a
    Prometheus counter, not a column)
ALTER TABLE devices ADD COLUMN device_connections_count bigint;
    -- → exit 1: connections? word-boundary

```

7.2 ALLOWED (build PASSES)

```

-- ALLOWED: identity/topology on the exhaustive allow-list (§5.3)
CREATE TABLE devices (
    id          uuid PRIMARY KEY,
    tenant_id   uuid NOT NULL,
    wg_pubkey   bytea NOT NULL,          -- ALLOW: identity (32B
        Curve25519 public key)
    overlay_ip  inet  NOT NULL,          -- ALLOW: identity
        (device's OWN stable overlay addr)
    last_seen_at timestamptz,           -- ALLOW: COARSE (≥5min
        coarsened, §2.2) – not a session timeline
    name        text
);
CREATE TABLE advertised_prefixes (
    connector_id uuid NOT NULL,
    cidr          cidr NOT NULL          -- ALLOW: topology (a
        connector's ADVERTISED network)
);

-- ALLOWED: control-action audit (the single ts+actor exception,
    meta shape-checked §9)
CREATE TABLE audit_events (
    id          bigint GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    tenant_id   uuid NOT NULL,
    actor        text NOT NULL,          -- a user id / "system" –
        never a destination
    action       text NOT NULL,          -- closed vocabulary (§4.1)
    target       text,
        -- opaque control-object id – never a destination
    ts           timestamptz NOT NULL DEFAULT now(),
        -- when the ADMIN acted (not a timestamp-of-use)
    meta         jsonb                  -- shape-guarded (§9)
);

-- ALLOWED: a comment mentioning a forbidden word (parser strips
    comments first, no false-fail)
CREATE TABLE handshake_totals ( -- aggregate only; no src_ip/
    dst_ip here
    day         date PRIMARY KEY,
    total        bigint
);

```

7.3 Edge-case classification table

Case	Verdict	Rationale
connections table added	FAIL	

Case	Verdict	Rationale
		direct S6/C3 breach (forbidden name)
device_connections_count bigint column	FAIL	connections? word-boundary; a legitimate aggregate must be a Prometheus counter, not a column
device_activity(remote inet, octets bigint, seen_at ts)	FAIL	combination rule (dst-shaped + bytes-shaped + ts) — a flow log dodging the name denylist
audit_events.meta containing {"src_ip":"..."} at runtime	FAIL	meta-shape guard (§9) — traffic-via-jsonb evasion
advertised_prefixes.cidr	PASS	topology allow-list
devices.overlay_ip	PASS	identity allow-list (the device's own addr)
view traffic_summary over only counts	FAIL	name is a forbidden signal; lint defaults to FAIL (safe); rename to metrics_summary
comment line mentioning bytes_out	PASS	parser strips SQL comments before matching (no false-fail on prose)

8. The event-payload lint (C3 on the bus)

The schema-lint guards the *durable store*. The **event bus** is the second place a traffic shape could leak in — an event payload could carry a src/dst pair just as a table column could. So C3 applies to the bus payload too, enforced by a mirror lint [svc-evt §2.2/§9.4, 04-sec §6].

8.1 The C3 payload constraint

Event payload objects are structurally constrained to **identity** / **topology** / **policy** keys. There is **no** field anywhere in the event taxonomy for a source/destination pair in use, a port a flow used, a byte count, or a DNS query [svc-evt §2.2]:

```
// The entire event payload taxonomy (svc-evt §3.4) – note: ZERO
// src/dst/bytes/dns fields.
type DeviceEnrolled struct { DeviceID uuid.UUID; Kind string;
    OverlayIP string } // overlay_ip = identity
type DevicePresence struct { DeviceID uuid.UUID; RTTMs
    uint32 } // rtt = health, not traffic
type DeviceRevoked struct { DeviceID uuid.UUID }
type ConnectorAttached struct { DeviceID uuid.UUID; Site string }
type ConnectorPrefixes struct { ConnectorID uuid.UUID; CIDRs
    []string; Gen int64 } // cidrs = topology
type RouteConflict struct { CIDR string; ConnectorIDs
    []uuid.UUID }
```

```

type PolicyVersion      struct { Version int64 }
type GatewayFailover    struct { From string; To string }

```

advertised_prefixes CIDRs and a device's stable overlay_ip *do* appear — they are **topology, not traffic** — exactly the same allow-list the schema-lint enforces [svc-evt §2.2]. The §9.4 payload-lint is the runtime signature that proves the constraint is live, not merely promised.

8.2 The bus payload-lint mirrors the schema-lint

The §1.1 payload-lint applies the same forbidden-field regex (§5) to event payload JSON shapes at build time (a test that introspects every payload struct) and at publish time as a cheap guard inside events.New [svc-evt §2.2/§9.4]. A payload struct that grows a dst_ip/bytes_in/ sni_host field FAILs the build — the bus-side analogue of the schema-lint. This closes the "log traffic via an event payload" evasion of C3.

9. The audit meta-shape guard (the jsonb back-door)

audit_events.meta is a jsonb blob — the one place a developer could smuggle a traffic field *inside* an otherwise-allowed audit row. The meta-shape guard closes that back door [svc-tel §4.4, §7.3].

```

// svc-tel audit.go (excerpt) – runs BEFORE every audit insert.
func (s *AuditSink) validate(r AuditRecord) error {
    if _, ok := auditVocabulary[r.Action]; !ok { return
        ErrUnknownAuditAction } // closed vocab §4.1
    if len(r.Meta) > maxMetaKeys { return
        ErrAuditMetaShape } // bound the blob
        (default 16 keys)
    for k := range r.Meta {
        if forbiddenMetaKey.MatchString(k) { return
            ErrAuditMetaShape } // §5 regex shared
        }
    }
    return nil
}

```

- The meta keys are matched against the **same** forbidden-column regex the schema-lint uses (§5.2). A control-action audit carrying a dst_ip/sni_host/ bytes_in key is **rejected** with ErrAuditMetaShape [svc-tel §4.4].
 - ErrAuditMetaShape **fails the control action closed** — audit is part of the action's transaction, so a refused audit rolls back the mutation it would have recorded [svc-tel §9]. Audit fails closed (accountability); presence fails static (never brick a tunnel) — the deliberate, load-bearing asymmetry [svc-tel §9].
 - The blob is bounded (default 16 keys) so audit cannot grow into a traffic log by accumulation.
-

10. Paired §1.1 mutation — proving the lint is not a tautology

A lint that passes everything is worse than no lint — it is a green light over a hole. So the no-logging gate is **self-validated** by a paired §1.1 mutation (§11.4.107(10) — an analyzer that passes its golden-bad fixture is itself a bluff gate) [04-sec §6.2, svc-tel §7.2].

10.1 The mutation contract

GOLDEN-BAD FIXTURE (the mutation):

Plant a throwaway migration:

```
CREATE TABLE flows (src inet, dst inet, bytes bigint, ts
timestampz);
```

ASSERT: schemalint exits 1 (FAIL) on this fixture. ← proves
the lint CATCHES a flow log.

GOLDEN-GOOD FIXTURE (the restore):

Remove the planted migration.

ASSERT: schemalint exits 0 (PASS). ← proves
the lint does not FAIL the clean schema.

VERDICT: the lint is real iff golden-bad FAILs AND golden-good
PASSes.

A lint that PASSes the golden-bad `flows` table is a tautology –
itself a §11.4 bluff gate.

10.2 The mutations (one per evasion vector)

Mutation (golden-bad)	Must produce
CREATE TABLE flows(src inet, dst inet, bytes bigint, ts timestampz)	lint FAIL (forbidden name + combination)
CREATE TABLE device_activity(remote inet, octets bigint, seen_at ts) (name-dodge)	lint FAIL (combination rule)
ALTER TABLE devices ADD COLUMN dst_ip inet	lint FAIL (forbidden column)
event payload struct grows a DstIP string field	payload-lint FAIL (§8)
audit_events.meta insert with key "src_ip"	ErrAuditMetaShape (§9)
strip the flows denylist entry from the lint regex	the mutation-test that asserts FAIL now PASSes ⇒ the <i>meta-test</i> FAILs (proves the denylist is load-bearing)

Each mutation must make its gate FAIL; restoring the clean state must make it PASS. This is the mechanical proof the no-logging invariant is *enforced*, not merely documented.

11. Runtime signature — green against the deployed DB (§11.4.108)

A green lint against the migration *source* proves the SOURCE layer only — the deployed DB could have drifted. The runtime signature (§11.4.108) requires the lint be green against the **live, deployed schema** [svc-tel §7.2, §10 T-CHAL-1].

- **The observable.** `schemalint` exit 0 against the *live* `information_schema` of the deployed Postgres is the runtime signature that S6/C3 is **active**, not merely promised [svc-tel §7.2].
- **Where it runs.** Not just pre-build — the §11.4.169 test matrix wires the lint as a **post-deploy** assertion against the deployed DB, so a drifted table that no migration mentions is caught on the running system, not just in CI [svc-tel §7.2, §10 T-CHAL-1].
- **The Challenge.** T-CHAL-1 drives a full enroll→policy→delta flow AND asserts `schemalint` green against the *deployed* DB, with captured evidence [svc-tel §10] — a Challenge that scores PASS on a privacy control that is actually broken is the same class of defect as a unit-test PASS-bluff.

This makes the no-logging guarantee a four-layer property (source / artifact / runtime-on-clean- target / user-visible per §11.4.108): the data does not exist in the source schema, does not exist in the deployed schema, and a passive observer correlating a user to a destination finds nothing because there is nothing to find.

12. Credentials discipline (§11.4.10) intersection

No-logging composes tightly with the §11.4.10 credentials mandate — both forbid sensitive data from landing where it can leak [04-sec §11, 04_ARCH §10, §11.4.10].

- **Secrets are never logged or printed.** `helixvpncctl` never prints or logs a credential, an enroll token, or an API token [04-sec §11]. The no-logging lint forbids *traffic* shapes; the §11.4.10 discipline forbids *secret* values in any log/output — complementary halves.
 - **Hashed at rest.** Enroll tokens are Argon2id-hashed (`enroll_tokens.token_hash`, plaintext never stored); API tokens hashed at rest [04-sec §2.2, §11]. A leaked DB dump yields no usable token and no traffic log.
 - **.env/secrets git-ignored project-wide** with a tracked `.example + README`, `chmod 600` on credential files [04-sec §11, §11.4.10/§11.4.30].
 - **The intersection invariant.** A debug log line that printed a destination would be *both* a no-logging violation (S6) *and*, if it also printed a token, a §11.4.10 violation — the two mandates overlap precisely at "no sensitive data in any durable or logged form."
-

13. GDPR / privacy posture

The data-minimization model (§2) *is* the GDPR posture: the strongest privacy guarantee is data that does not exist [04-sec §6, §7].

GDPR concern	HelixVPN posture	Mechanism
Data minimization (Art. 5(1)(c))	only identity/topology/policy + control-action audit is stored; no traffic, no destinations, no DNS, no timestamps-of-use	§2 model + §6 schema-lint enforce it mechanically
Purpose limitation (Art. 5(1)(b))	audit records <i>control actions</i> (operate the service), never user traffic	§4 closed vocabulary + §9 meta guard
Anonymous mode (no PII)	a tenant can mint device enroll tokens with no email, no SSO — the Mullvad "account number, no PII" stance; <code>users.email = NULL, oidc_sub = NULL</code> , no reverse-link to a human	[04-sec §2.2]
Right to erasure (Art. 17)	identity/device data is deletable; there is no traffic log to erase because none exists; <code>helixvpnctl audit prune --before <ts></code> exists for operator-driven audit erasure	[svc-tel §4.1] (operator-driven, not automatic)
Storage limitation (Art. 5(1)(e))	presence is ephemeral (TTL, §3); <code>last_seen_at</code> is coarse (§2.2); audit is low-volume control-only with optional operator prune	§3 + §2.2 + [svc-tel §4.1]
Subject-access (Art. 15)	what HelixVPN holds about a device is enumerable (identity/topology rows) and provably bounded by the schema-lint — there is no hidden traffic store to disclose	§2 + §6

Honest boundary (§11.4.6). No-logging-as-code guarantees the *durable store and the bus and the audit blob* hold no traffic data — it does **not** claim a hot-compromised gateway holds nothing in RAM. Live in-RAM presence on a compromised running gateway, and aggregate counters, are the residual surface (§14, threat T5). The guarantee is "no durable correlation of a user to a destination ever exists," not "an attacker with root on a running box sees nothing in memory."

14. Threat-model placement & residual risk

No-logging-as-code closes threat **T5** in [04-sec §10]:

#	Threat (STRIDE)	Vector	Mitigation (this doc)	Residual risk (§11.4.6)
T5				

#	Threat (STRIDE)	Vector	Mitigation (this doc)	Residual risk (§11.4.6)
	Information disclosure — traffic correlation	Operator/host compromise reads a connection log	No connection log exists (S6, §2) + CI schema-lint (§6) + ephemeral Redis presence (§3)	Live in-RAM presence on a <i>hot-compromised</i> gateway; aggregate counters only — no durable user [?] destination record to read

Residual risks stated honestly:

1. **Hot-compromise RAM.** A gateway compromised *while running* has live presence keys in Redis RAM (who is online now) and in-flight state. This is ephemeral and carries no destination, but it is not nothing. The guarantee is *durable* absence, not *runtime-memory* absence.
2. **Aggregate counters.** Prometheus series (handshakes, bytes-total, errors) exist by design — they are aggregate, label-cardinality-bounded (no tenant_id/device_id/ip labels, §3.1 of svc-tel), and cannot correlate a user to a destination. But they reveal *operational shape* (tenant population via gauges), which is why /metrics is scrape-network-only (§8.3 svc-tel).
3. **Lint-evasion creativity.** The denylist (§5) is closed; a sufficiently creative column name could in principle dodge it. The combination rule (src/dst + bytes/ts) catches name-dodges, the live-schema cross-check catches drift, and the paired mutation (§10) keeps the denylist load-bearing — but a reviewer (§11.4.6) is the backstop for a genuinely novel shape, and the lint defaults to FAIL (safe) on an ambiguous name.
4. **Not a runtime-memory claim.** Per §13's honest boundary — no-logging is a durable-store + bus + audit guarantee, not a claim about an attacker with root on a live box.

15. Phase → task → subtask plan (no-logging workstream)

Each leaf becomes a §11.4.93 workable item. The no-logging lint + RLS audit are the **irreversible privacy/correctness floor and land first** (§11.4.132 risk-descending) [svc-tel §11]. Anti-bluff: every closure carries captured evidence (§11.4.5/§11.4.69) and a paired §1.1 mutation.

Phase 1 (MVP)

- **NL-1 — schema-lint gate.** tools/schemalint: static migrations + live information_schema parse; forbidden-table regex (§5.1); forbidden-column regex (§5.2); exhaustive allow-list (§5.3); combination rule (src/dst + bytes/ts); comment-stripping; verdict logic (§6.1). Wire into pre-build [svc-tel TEL-T1, 04-sec §6.2].

- **NL-2 — paired §1.1 mutation.** Plant flows(`src,dst,bytes,ts`) $\Rightarrow$ lint FAIL; remove $\Rightarrow$ PASS; the name-dodge device_activity $\Rightarrow$ FAIL; golden-good/golden-bad self-validation (§10) [svc-tel T-LINT-1].
- **NL-3 — ephemeral presence.** PresenceTracker Redis SET/SADD/Expire/Del with PRESENCE_TTL = 3×HEARTBEAT; keyspace-expiry reaper + PRESENCE_SWEEP sweeper; never copy to Postgres; coarse last_seen_at (≥ 5 min) (§3) [svc-tel TEL-T3].
- **NL-4 — control-action audit.** audit_events closed vocabulary (§4.1); append-only by grant (REVOKE UPDATE, DELETE); meta-shape guard (§9); WithTenant RLS insert; fails-closed (§9) [svc-tel TEL-T2, 04-sec §7].
- **NL-5 — event-payload lint.** Mirror the schema-lint over event payload structs (§8); build-time introspection + events.New publish-time guard [svc-evt §9.4].
- **NL-6 — runtime signature.** Wire schemalint as a post-deploy assertion against the deployed DB (§11); T-CHAL-1 end-to-end Challenge asserting deployed-DB lint green [svc-tel §7.2, §10].
- **NL-7 — credentials intersection.** Argon2id token hashing; helixvpncctl never prints secrets; .env/secrets git-ignored (§12, §11.4.10/§11.4.30) [04-sec §11].

Phase 2 (parity + hardening)

- **NL-2.1 — lint holds across phases.** The schema-lint gates the Phase-2 NATS-JetStream swap and the optional ClickHouse aggregate sink (an *export of already-aggregated counters*, never a connection log — C3 holds across phases, lint gates Phase 2 too) [svc-tel §12].
- **NL-2.2 — GDPR tooling.** Subject-access enumeration (§13 Art. 15) + operator-driven helixvpncctl audit prune (§13 Art. 17) hardened with captured-evidence tests.

Sources verified

- `final/04-security-privacy-pki.md` ([04-sec]) — §0.1 invariants **S6** (no durable connection/traffic log by construction) + **S7** (control actions audited, traffic never); §6 (no-logging as code: §6.1 stored-vs-not-stored table, §6.2 CI schema-lint Go code + anti-bluff paired mutation); §7 (audit control-actions only, closed action set, append-only, live stream); §2.2 (anonymous enroll tokens, no PII, Argon2id hash); §10 threat **T5** (traffic correlation) + residual risk; §11 (secrets hygiene, §11.4.10/§11.4.30); §13 S6/S7 verification rows. — local spec.
- `v03-control-plane/svc-telemetry.md` ([svc-tel]) — §0 (telemetry owns the no-logging-as-code invariant + governs tools/schemalint); §4 (control-action audit: §4.1 DDL append-only-by-grant, §4.2 closed AuditAction vocabulary, §4.4 meta-shape guard); §5 (live presence in TTL Redis: §5.1 key schema + PRESENCE_TTL = 3×HEARTBEAT + coarse last_seen_at, §5.3 expiry→offline event
 - sweeper, §5.4 drift handling); §7 (the schema-lint nano-detail: §7.1 forbidden-table/column regex + exhaustive allow-list, §7.2 verdict logic + runtime signature + paired mutation, §7.3 edge-case table); §9 (error taxonomy — audit fails closed, presence fails static); §10 (T-LINT-1 / T-CHAL-1 / T-STORE-1 test points); §11 (risk-descending task order — lint

- + RLS audit first); §3.1 (metric label cardinality bound — no tenant/device/ip labels); §12 (lint gates Phase-2 NATS/ClickHouse). — local spec.
- `v03-control-plane/svc-events.md` (`[svc-evtl]`) — C3 (no-logging by construction on the bus); §2.2 (Envelope privacy boundary — payload is identity/topology/policy only, no src/dst/bytes/dns field; `overlay_ip`/CIDRs are topology not traffic; §9.4 payload-lint as runtime signature); §3.4 (full payload taxonomy — zero traffic fields). — local spec.
- [`04_P1` §11.4] HelixVPN-Phase1-MVP — no-logging CI lint + DoD; [`04_ARCH` §7] no-logging-as-code
 - control-only audit; [`04_ARCH` §4.5/§4.6] ephemeral Redis presence operationalizes no-logging.

*Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — every unproven claim marked **UNVERIFIED**; presence TTL derived from heartbeat, never invented; lint defaults to FAIL on ambiguous shape), §11.4.10/§11.4.30 (credentials/secrets hygiene intersection, §12), §11.4.107(10) (self-validated analyzer — golden-bad flows table ⇒ lint FAILs, §10), §11.4.108 (runtime signature — lint green against the DEPLOYED DB, not source grep, §11), §11.4.132 (risk-descending — privacy floor lands first, §15), §11.4.144 (presence expiry never a silent corpus hole, §3.2), §11.4.93 (phases → workable items), §11.4.169 (post-deploy test-type matrix wires the runtime signature), §11.4.65/§11.4.153 (HTML+PDF[+DOCX] exports follow in refinement). Honest boundary (§11.4.6): no-logging is a durable-store + bus + audit guarantee, NOT a claim about runtime memory on a hot-compromised gateway (§13/§14 residual risk T5).*